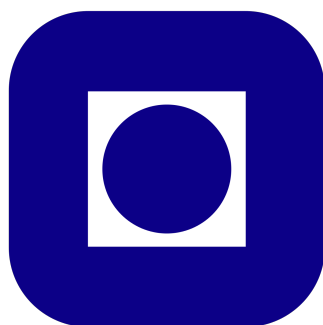


NTNU



INTELLIGENT TEXT ANALYTICS AND LANGUAGE
UNDERSTANDING

TDT4310

Project Plan

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1 Paraphrase Detection

Given a pair of sentences, Paraphrase detection is the task of determining if the two sentences convey the same meaning. Paraphrase detection can be useful in various natural language processing tasks, such as information retrieval, information extraction and text simplification for improving readability.

1.1 Training data

Data Source: The training data for this project will be retrieved from the Quora Question Pairs competition hosted on Kaggle. The competition dataset consists of a collection of question pairs (400K pairs) labeled as duplicate or non-duplicate, and it is available for download from the following link: [Quora Question Pairs Competition Data](#).

1.2 Methods

The main envisioned method to tackle this kind of task is a Transformer-based model, specifically ALBERT [1] (A Lite BERT for Self-supervised Learning of Language Representations). Compared to BERT, ALBERT is a smaller model containing less parameters. Fine-tuning ALBERT involves adjusting its parameters using task-specific data to optimize its performance for the task at hand, in this case, paraphrase detection. This process typically involves modifying the last layers of the ALBERT architecture and training the model on paraphrase pairs to learn how to detect if two sentences are paraphrases of each other. The model is available at the following link: [ALBERT](#). In order to evaluate the performance of the model, its predictions from the training dataset will be compared with labels present in the testing dataset to assess the model's performance, with fine-tuned hyper parameters for validation. Additionally, a SVM (support vector machine) classifier will be trained from scratch and used as baseline to compare the results obtained from the fine-tuned ALBERT model.

References

- [1] Zhenzhong Lan et al. *A Lite BERT for Self-supervised Learning of Language Representations*. 2020. URL: <https://arxiv.org/pdf/1909.11942.pdf>.